

PSK-2 CNC Logging System



PSK-2 CNC Logging System is a digital logging system specially designed for the convenience of field work, which can connect with various kinds of pipe and log. Because this system uses a series of advanced technology, it improves the use of one ground-based instrument controlling one heavy logging probe into one ground-based instrument controlling various kinds of logging probe, it also includes the deep measurement, the power supply and so on. The instrument has a light weight and simple operation and maintenance, which can be connected to the tube type, which can be aseismic, temperature-resistant, moisture proof and high reliability.

Description

PSK-2 CNC Logging System integrated digital control logging system is a new generation of digital logging equipment, which based on summarizing more than 20 years in production and successful experience of digital logging equipment's research and development, following the international new trend of development of the small digital logging technology, and combining with China's national conditions. This set of equipment has the following advantages over other similar products:

- The measurement parameters are large and the scope is wide.
- The reliability is high and the Mean Time Between Failure (MTBF) is long.
- Portable, convenient for mobile transportation.
- Operating and maintenance is simple.

The main reason for these advantages is that the design and development is absorbing other's merits, fully considering the various needs of the industry, and considering the competitiveness in the international market PSK-2 CNC system can measure than 20 kinds parameters, covered with metal ore, coal, oil field, radioactive hydrology, and other areas of the various user's requirements, with great cost-effective.

Features & Benefits

1. Standardization and modularity

In the PSK-2 CNC integrated digital control logging system, both hardware and software, no matter probe or ground box, signal measurement or signal transmission, have adopted the standardization and modularization design concept. The benefits of standardization and modularity Product is excellent in products' consistency, maintainability and reliability. It is convenient to operate and maintain. The accessories are 100 percent universal and can be expanded at any time.

PSK-2 CNC Logging System

2. Fully underground digital and measurement's signal transmission

All the signals from the underground probe are converted by A/D and sent to the ground after being encoded.

The underground probe is a two-way street communication. Digital code transmission measurements greatly improve the anti-jamming capability, and the system itself can correct the error code, which guarantees the quality of the signal transmission.

3. Oversampling technology to obtain the most formation information

The JHQ-2D system uses the sampling method for the collection of formation information. The oversampling amount increases with the decrease of the logging speed, and over sampling ensures the need for digital filtering, enhancing spatial resolution and screening useful information.

4. Small size of the underground probe with high pressure

PSK-2 CNC probe is almost diameter $\Phi 40$ and $\Phi 50$ mm, the appearance of length 1000~1500mm size. The probe is made of stainless steel. Resistance to 20~40 MPa water pressure, temperature of 70 ~ 90°C. Special orders can be 60 MPa water pressure resistance, temperature 120°C resistance.

Parameter

1. The use of the environment

- Power supply: 220V, 50 Hz;
- Using the environment temperature: -20°C ~ 45°C;
- Using the environment humidity: 90% or less (45°C).

2.The depth measurement system

- Depth measurement range: 0 ~ 9999.99 m;
- Depth resolution: 0.28 mm;
- Speed measurement range: 0 ~ 9999m/h;
- Deep optical pulse signal: 3600 pulses per meter.

3.Signal transmission system

- Bottom hole output signal: bipolar coding;
- Communication with the computer: RS232 or USB;
- The underground signal detection level: plus or minus 0.5 V ~ plus or minus 12V.

4. Figure out the depth of the graph

- 1:10;
- 1:50. 1:100. 1:0;
- 1, 50, 0, etc.

5. The drawing accuracy

- 8 points per millimeter (curve);
- 64 pixels per square millimeter (image method).

GESL-2Dr Natural Gamma & Electrical Resistivity probe



Description

The Electrical Resistivity probe measures natural gamma, gradient resistivity, normal resistivity and spontaneous potential (SP) simultaneous GESLy. Users can log resistivity profiles with different depths of investigation and gain information about permeability, porosity, water quality and geological formation properties.

Operating Conditions: Uncased; Borehole Fluid: Water or Mud.

Features & Benefits

1. The probe will digitize the measured information of the stratum and transmit it to the surface to avoid the effect of the cable on the measurement data.
2. Constant power, adaptive measurement, no manual operation.

Applications

- Facies changes; Quantitative geological formation properties
- Identification of hydrostratigraphic units
- Aquifer thickness; Water quality
- Identification of hydrocarbon intervals
- Detection of ore body zones; Bed boundary analysis

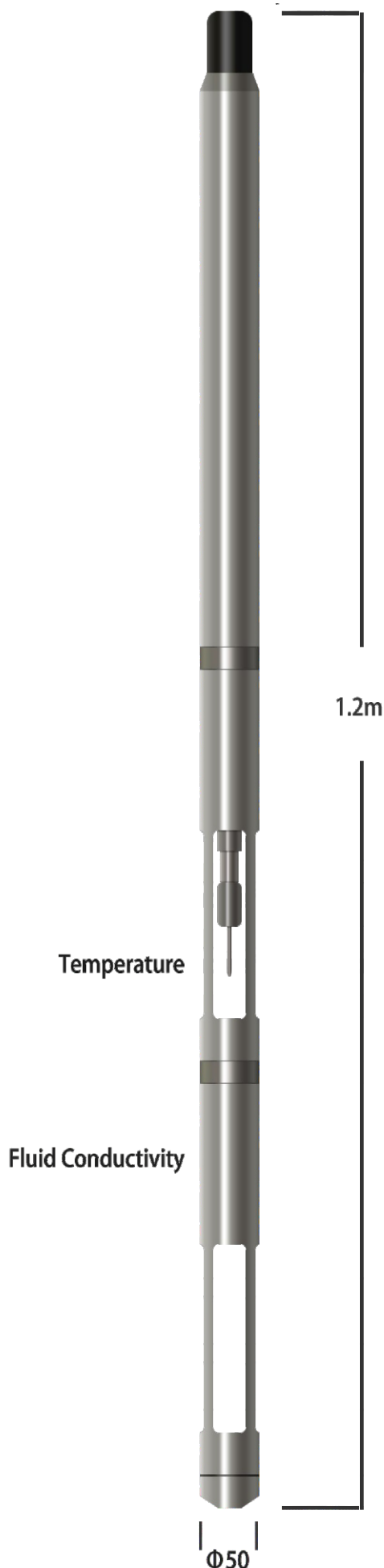
Parameter

1. Instrument power supply: DC200V $\pm$ 20%, Electricity $\leq$ 40mA
2. Electrode array: N 0.6 M2 1.3 M1 0.3 A
3. Natural gamma sensor: NaI Crystal
4. Measuring range:
 - Apparent resistivity: 1 ~ 4000 Ω m (customize 10000 Ω m)
 - Spontaneous potential (SP): $\pm$ 1200mV
 - Natural gamma: 0 ~ 32768cps
5. Measurement accuracy: 2% (10 ~ 4000 Ω m) or $\pm$ 1 Ω m
6. Environmental: Temperature $\leq$ 80 $^{\circ}$ C; Pressure $\leq$ 20MPa

Related Products

- Natural Gamma & Electrical Resistivity probe
- Electrical Polarizability Probe
- Simulated Graduator
- Isolation Electrode

GESL-1 Temperature & Fluid Conductivity Probe



Description

The Fluid Conductivity and Temperature probe provides borehole temperature and fluid conductivity measurements. The borehole temperature measurement is helpful to detect anomalies caused by events such as fluid flow into the borehole. The borehole fluid conductivity is directly proportional to the concentration of dissolved minerals. It is generally used in hydrogeology to determine the concentration of dissolved ions in the aquifers and to locate the fluid flows occurring in the borehole.

Operating Conditions: Uncased; Borehole Fluid: Water or Mud.

Features & Benefits

1. The sensor adopts pt100 platinum resistance of a small time constant and can quickly identify temperature changes in the stratum.
2. Strong resistance to jamming.

Applications

- Fluid conductivity – salinity
- Salt-water intrusion studies
- Identification of fluid flow in open/cased hole
- Localization of the water table
- Water-well monitoring
- Localization of water intervals of different quality
- Geothermal gradient logging
- Often used in the implementation of temperature compensation equations for other logs

Parameter

1. Temperature Range: 0°C ~ 100°C, Resolution: 0.025°C
2. Temperature Sensor: Platinum Resistance (pt100)
3. Conductivity Range: 0 ~ 200Ωm, Resolution: 0.05Ωm
4. Electrode array: A 0.04 M 0.02 N 0.02 N, ∞ B
5. Signal: bipolar codes
6. Environmental: Temperature ≤ 80°C; Pressure ≤ 20MPa

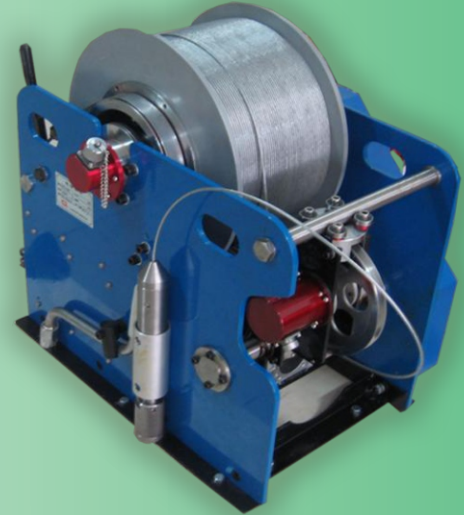
Related Products

- Fluid Conductivity and Temperature probe
- Temperature probe
- Borehole Deviation and Temperature probe

DJ Winch

DJ series winch is a special equipment for geophysical exploration and geological survey. It is mainly used with integrated digital logging equipment and can also be used with other logging equipment. Most DJ series winches are anti-magnetic design, stepless speed adjustment and automatic cable arrangement. The performance can be applied to various geological prospecting areas.

DJ series winch equipment adopts frequency conversion motor, dc motor, ac asynchronous motor as power source, and has a wide range of speed regulation, smooth variable speed, convenient operation, simple maintenance and so on. It can be used in geophysical prospecting and geological survey work on the land, sea, plateau (which is less than 5,500 meters above sea level) and other environments on the earth.



Features & Benefits

1. Standardization and modularity

DJ series logging cable winch is powered by marine frequency conversion motor, permanent magnet DC motor, industrial frequency conversion motor, dc motor, ac asynchronous motor as power source.

The selection of power source mainly considers the factors such as high temperature and humidity environment, high altitude environment, convenient field transportation, convenient maintenance, long-term continuous operation and so on, so as to ensure the reliability of the whole set of equipment.

2. Automatic cable arrangement function

Most DJ series logging cable winch has automatic cable arrangement function, which ensures that the cable can be precisely and neatly arranged when dropping and lifting, and can greatly prolong the service life of the measuring cable. It is very important that the winch has automatic cable arrangement function especially for steel wire armored cable.

3. Standardization and modularity

DJ series logging cable winch has the automatic braking function. The winch can automatically stop when encountered in the use process of emergencies, such as power outages, winch controller failure and so on to keep the tube safe, so the tube will not fall.

4. Manual function

The DJ series logging cable winch is equipped with a handle, which can drop or lift by hand without electricity in special circumstances.

5. Accurate measurement of cable length

The DJ series logging cable winch has precise cable metering device to ensure that the instrument has an accurate depth in the process of lifting or dropping of cables.

DJ Winch

6. Resistance to salt fog

Main components of DJ series cable winch for Marine geophysical survey adopt high strength aluminum alloy as the material. The surface is coated with electrostatic spraying, some fasteners are stainless steel. It takes Marine ac motor as power source, and the cable has kevlar wire and wire twisted rope core and it is coated with polyurethane.

7. The magnetization

For the purpose of electric logging, magnetic logging and other projects, the main structure of the DJ series logging cable winch adopt high strength aluminum alloy as material, which ensures that the logging cable can not be magnetized, and keep the minimum amount of self and mutual inductance.

Parameter

1. Cable length measurement: automatic measurement
2. Photoelectric pulse generator: SG9208, 1800P/R for each transfer pulse
3. Transmission device: JLH0710
4. Self-locking brake and manual brake
5. Class of electrical motor insulation: Grade F
6. Shell protection level: IP56
7. Use of the environment:
 - less than 5500 meters, ambient temperature - 20°C ~ + 45°C, relative humidity 99% or less
 - Power: 220V current 5A



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